Research Project: Franciscan Eco-Theology and Environmental Attitudes

A term paper submitted for the
RM in Management and Applied Microeconomics

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Abstract

We analyse the effect of exposure to two Franciscan and Dominican Mendicant orders between the 13th and 15th centuries on contemporary environmental attitudes. We find that higher Franciscan exposure is associated with less pro-environmental attitudes, contrary to our initial expectations, whereas the effect of Dominican exposure is statistically insignificant. We suggest that these results could be driven by differences in orientation toward collective action. This possible underlying mechanism is consistent with the contrasting theologies of the two orders regarding collectivity and individualism.

1 Introduction

The Catholic Church, played an important role in medieval Europe, shaping people's lives, values, and behaviour throughout the Middle Ages (Lynch, 2014). One of the most famous religious figures of the Middle Ages is St. Francis of Assisi, founder of the Franciscan Order and patron saint of ecology within the Catholic Church, widely known for his deep affection towards nature and animals.

We investigate what is the effect of exposure to the Franciscan "Eco-Theology" between the 13th and 15th centuries on contemporary environmental attitudes among Europeans. We use as a control the exposure to the Dominican order, which was another major Mendicant Order during the Middle Ages.

To do so, we construct historical exposure measures for both Franciscan and Dominican Orders as independent variables, defined as the fraction of a region's historical population living geographically close to a monastic house. We then construct our dependent variable, the environmental index, via Principal Component Analysis to individual-level survey data on environmental attitudes from the European Values Study (EVS). Using those variables and a set of individual-level controls, we empirically examine how historical exposure to the two Mendicant Orders affects contemporary environmental attitudes.

Given the historical background of St. Francis of Assisi, we have set the following initial hypothesis: higher historical exposure to the Franciscan Order is associated with more pro-environmental attitudes, while higher exposure to the Dominican Order is associated with less pro-environmental attitudes. However, contrary to our expectations, we find that higher exposure to the Franciscan Order is associated with less pro-environmental attitudes, while exposure to the Dominican Order had statistically insignificant results.

We propose a possible mechanism that might have driven those results, grounded on the underlying structural differences of both Catholic Orders on the collective action and individualism. The Franciscan theology of humility and acceptance of circumstances might have led to less pro-environmental attitudes, while the Dominican theology of persuasion and institutional coordination might have led to more pro-environmental attitudes.

2 Historical Context and Hypothesis

At a glance, both Franciscans and Dominicans were Mendicant Orders, meaning that they relied on charity and begging for their livelihood and were more committed to preaching and teaching than owning land or monastic seclusion. Founded in the same century, both orders were operating as branches of the Catholic Church and operated widespread across Europe, with significant influence on the medieval society. However, despite having these similarities, the two orders differ in their theological emphasis and approaches (Arruñada and Lopez-Manuel, 2024).

The Franciscan Order was founded in the early 13th century by St. Francis of Assisi (1181-1226), who was recognized by the Catholic Church as the patron saint of ecology, widely known for his profound affection towards nature and animals, as seen in his "Canticles of the Creatures," where he refers to sun and moon as brother and sister. Francis, together with his followers, emphasized the close imitation of Christ and the apostles through poverty and humility. "*He wanted to embrace real poverty, without any rationalisations* *or modifications. His band of followers would beg for their daily food, would take no care for tomorrow, and would own nothing – really nothing.*" (Lynch, 2014, p. 258).

In contrast, the Dominican Order, founded by St. Dominic (1170-1221) in the early 13th century, prioritized intellectual and educational mission of medieval Catholic Church. "*For Dominic, a life of poverty and begging was a means to an end, whereas for Francis, it was an end in itself*" (Lynch, 2014, p. 262). Recognizing that learning was important to train preachers, the Dominican Order had internal schools (Lynch, 2014). They also founded universities, and produced extensive theological and practical writings (Le Goff, 1980; Boyle, 1981).

In short, Franciscans focus on collective life, communal care, antimaterialism, shame, compassion, simplicity, and humbleness (Arrunada and Lopez-Manuel, 2024). In contrast, given the nature of their activities, Dominicans emphasized critical thinking, self-examination, and moral development (Hinnebusch 1966; Lesnick 1989).

Because of the strong association of the Franciscan Order with nature and animals, we hypothesized that higher historical exposure to the Franciscan Order would be associated with more pro-environmental attitudes, while higher historical exposure to the Dominican Order would be associated with less pro-environmental attitudes.

3 Literature Review

Similar to our approach, Arruñada and López-Manuel (2024) examined the effects of the two major Mendicant Orders of the Middle Ages, the Dominican and the Franciscan, on the traits of the interpersonal exchange. Through their emphasis on guilt, shame, compassion, and analytical thinking, these orders shaped cognitive, interpersonal, and institutional outcomes, which together constitute the "foundations" of impersonal exchange. They further argue that eco-theological beliefs of both Mendicant Orders "*differ sharply in their approach to both spiritual and mundane life" and can foster distinct values and cultural norms*" (Arruñada and López-Manuel, 2024, p. 2).

For example, at the interpersonal level, "Dominicans promoted guilt as a driver of prosocial behaviour toward strangers", fostering a culture of individual accountability, independence, generalized fairness and trust, "*whereas Franciscans encouraged compassion and shame, which strengthened bonds within the group but limited cooperation beyond it*" (Arruñada and López-Manuel, 2024, p. 4).

Consistent with these differences, the authors document contrasting effects of Dominican and Franciscan influence on impersonal exchange. Their findings are consistent with our results regarding the contrasting effects of the two orders, as well as our proposed mechanism that the underlying differences in theological beliefs might have driven the results.

Our study also relates to the broader literature on the role of medieval churches and persistence of cultural norms. The Catholic Church played an influential role in every aspect of life in medieval Europe, shaping people's lives, values, and behaviour throughout the Middle Ages (Lynch, 2024). Once a religious institution establishes in a community, its values and norms tend to persist as cultural legacies, embedded in national cultures and are transmitted through educational institutions, even after losing some of the formal allegiance of their followers in modern Europe (Inglehart and Baker, 2000).

Another important strand of literature relates to the determinants of environmental attitudes. It has been examined how cultural differences might influence environmental attitudes (Chwialkowska et al., 2020; Khan et al., 2024; Yang et al., 2024; Tam, 2025, as cited in Behler, 2025). Some literature also suggests that religiously grounded nature world-view is associated with how people understand and relate to the environment (Cohen et al., 1976).

On top of that, some theoretical models argue that pro-environmental attitudes may be rooted in self-interest and utility maximization. In particular, a longer life expectancy may increase individuals' willingness to support environmental efforts to fight climate change (Mariani et al.,2010). Additionally, importantly for our research, many other studies with similar interests commonly measure the environmental attitudes using survey datasets, such as the EVS (Gelissen, 2007; Behler, 2025).

Taken together, these strands of literature support and motivate our research, which examines the effect of historical exposure to Franciscan and Dominican monasteries on contemporary environmental attitudes. Now the task is to come forward with a methodological strategy to isolate the effects previously discussed.

4 Data and Methodology

4.1 Exposure Measure

In our reseach, we assume that people who are living close to a monastery get influenced or exposed to its theological beliefs. Thus, the exposure measure is defined as the fraction of a region's historical population living geographically close to a monastic house. To estimate historical exposure to the Franciscan and Dominican Orders, we employ the geodataset covering the years between 1216 to 1500 from the Mapping Past Societies (MAPS), which includes information on the geographic locations of the Dominican monasteries consisting of 870 observations and Franciscan monasteries consisting of 994 observations. The MAPS dataset reports foundation centuries for Dominican monasteries only, but not for Franciscan monasteries. Due to the lack of this information, we assume that all monasteries in our dataset existed throughout the period of 1200-1500 since the majority of the monasteries were founded in the 13th century. However, this is also a major limitation of our approach, since it means that we are not able to capture the changes in the exposure over time, and we are only able to capture the overall exposure level. After we get the monastery location, we need to locate them within regions. To do so, we use the European Commision system for characterising geographical data, the Nomenclature of Territorial Units for Statistics (NUTS), as well as country shapefiles. NUTS-1 correspond large macro regions, such as German federal states, while NUTS-2 are more localized regions within countries. Since Germany has modified its NUTS-2 administrative borders during the collection period of the EVS data, we rely only on the larger NUTS-1 regions for Germany, and use NUTS-2 for all other countries. We additionally needed to delete monasteries in east Ukraine, Crimea, and Middle Eastern cities (Jerusalem/Acre), since those monasteries lie in locations that are not in or close to any NUTS region. There are some monasteries, however, that are outside but still close enough to a NUTS region, so we coerce them to fit into the closest region. As seen in Figure 1, the two Ukrainian monasteries were coerced to Eastern Romania, since they are pretty close to it. Some Turkish monasteries appear in the dataset because the EU also collects data on parts of Turkey, although these observations are not included in the analysis (for lack of matching environmental attitudes data). They do not affect the results or procedure and we retain these observations for completeness. Similarly, we can also see monasteries in Greece, but those should not affect the quantitative approach, since there are no entries for Greece in the EVS environmental survey data.

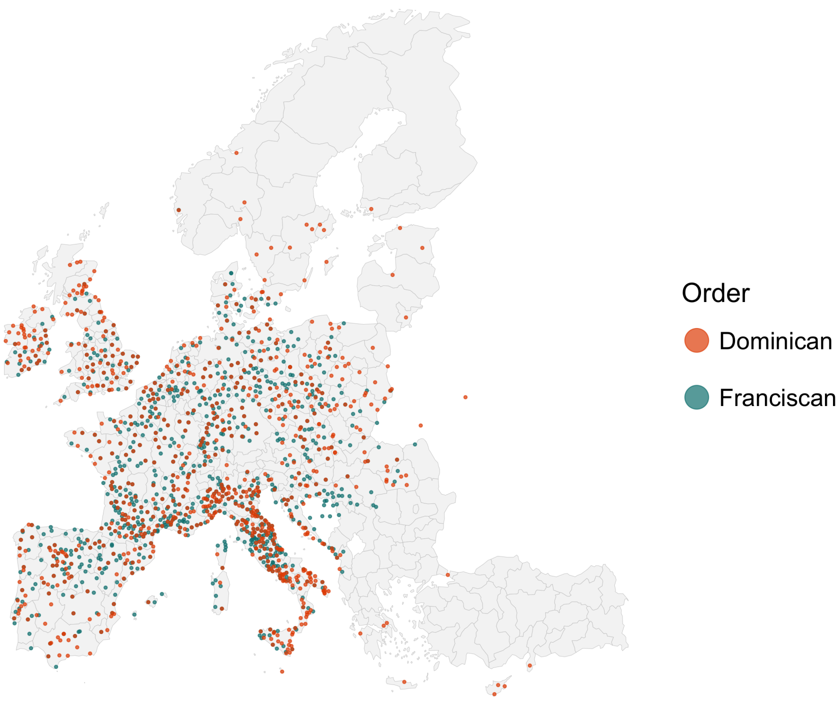
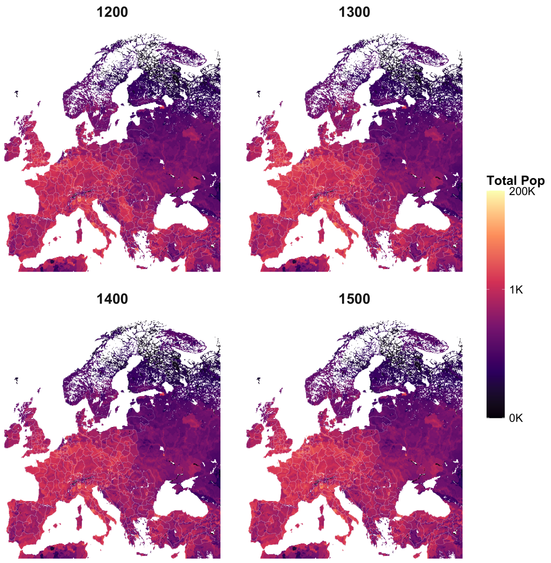


Figure 1: Locations of Franciscan and Dominican Monasteries

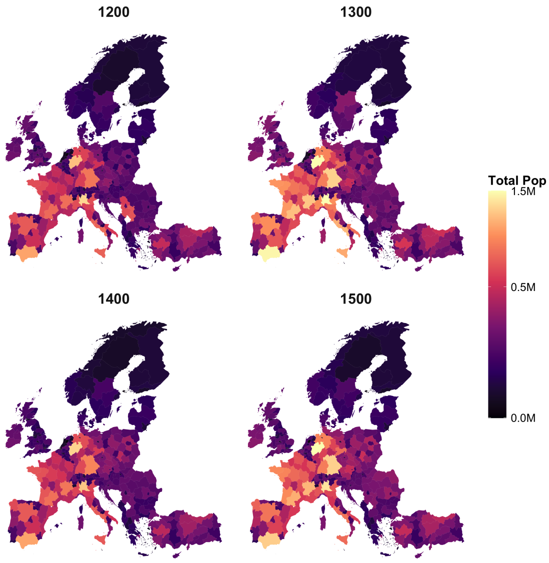
Once we have combined the monastery locations with NUTS shapefiles, our next step is to observe historical population data.

HYDE 3.3 (History Database of the Global Environment) Utrecht University population data captures where people lived historically. It provides estimates of population counts and density on a regular latitude-longitude grid from the very late medieval time to the modern period. Grids are squares of 85 km^2, which is quite granular, and we have population estimates for each grid. In Figure 2(a), each cell represents estimated population counts for a historical year (1200–1500).

Based on the population densities in grid squares, we construct population estimates at the NUTS level. In Figure 2(b), population data from HYDE 3.3 grids are overlaid with NUTS boundaries where each grid cell is assigned to a modern NUTS region and the total population within a NUTS region is calculated as the sum of the population of all grid cells that fall within that region. If a grid cell is partially within a NUTS region, we assign the population of that grid cell to the NUTS region if more than half of the grid cell's area falls within the region. This way, we can estimate the population of each NUTS region based on the HYDE 3.3 grid data.



(a) Population densities (grid-level, log scale)



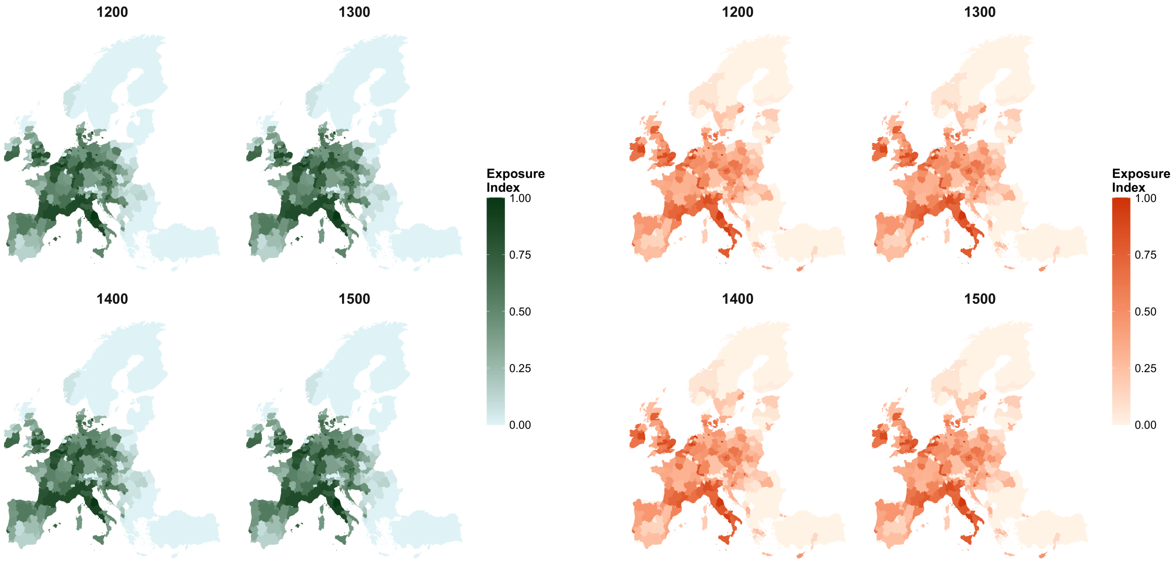
(b) Regional populations (NUTS-2 level)

Figure 2: Population measures from HYDE 3.3

Finally, we match the population estimates with the monasteries locations. We draw a 25 km radius circle around each monastery, and all the grids that are inside that circle are counted as the exposed population. All grids which have more than half its area included in the radius are counted as fully exposed to the theology of that monastery. In our research, the exposure measure is defined as the sum of exposed population within a given region as a percentage of the total population for both Franciscan and Dominican monasteries:

$$\text{Exposure} = \frac{\text{Exposed Population}}{\text{Total Population}} \tag{1}$$

With all three data sources combined, we have Dominican and Franciscan exposure measures for Europe (Figures 3(a) and 3(b)).



(a) Franciscan exposure (b) Dominican exposure

Figure 3: Map of exposure to Mendicant Orders in Europe (1200–1500)
normalized index (0 = none, 1 = high exposure)

4.2 Environmental Attitudes Measure

To construct our main dependent variable, the environmental index, we use the European Values Study (EVS) 2017–2021 dataset, which is individual-level survey data on values, beliefs, attitudes, and socioeconomic characteristics across European countries. The dataset covers over 30 European countries, of which 24 are relevant for our research when we exclude the countries that are not in NUTS-2. After data cleaning, the final dataset includes 49233 observations and geographic identifiers for NUTS-1 regions for Germany and NUTS-2 for all other countries.

From the EVS dataset, we select six environmental variables in particular, which correspond to questions related to environmental attitudes. The questions are the following:

Survey Question (EVS)	**Dimension**
Choosing between "protecting the environment priority, even if slower economic growth and loss of jobs" vs. "economic growth and creating jobs priority, even if environment suffers"	Economy–environment trade-off
"I would give part of my income if I were certain that the money would be used to prevent environmental pollution."	Willingness to pay
"It is just too difficult for someone like me to do much about the environment."	Self-efficacy
"There are more important things to do in life than protect the environment."	Priority of environmental protection
"There is no point in doing what I can for the environment unless others do the same."	Collective action
"Many of the claims about environmental threats are exaggerated."	Env. skepticism

The questions have categorical answers ("agree strongly", "agree", "neither agree nor disagree", "disagree", "disagree strongly", or choosing between the two statements). However, the questions differ in their directional interpretation. Thus, we need to standardize the questions by assigning a 0-1 scale for each answer to each question (see Table 2 in the Appendix), such that the answers indicate the same environmental direction. In the standardized 0-1 scale, 1 indicates the least pro-environmental attitude, and 0 indicates the most pro-environmental attitude. See Figure 4 for visualization for environmental questions within the NUTS regions. The green regions indicate being more pro-environmental, while the yellow regions indicate less pro-environmental attitudes.

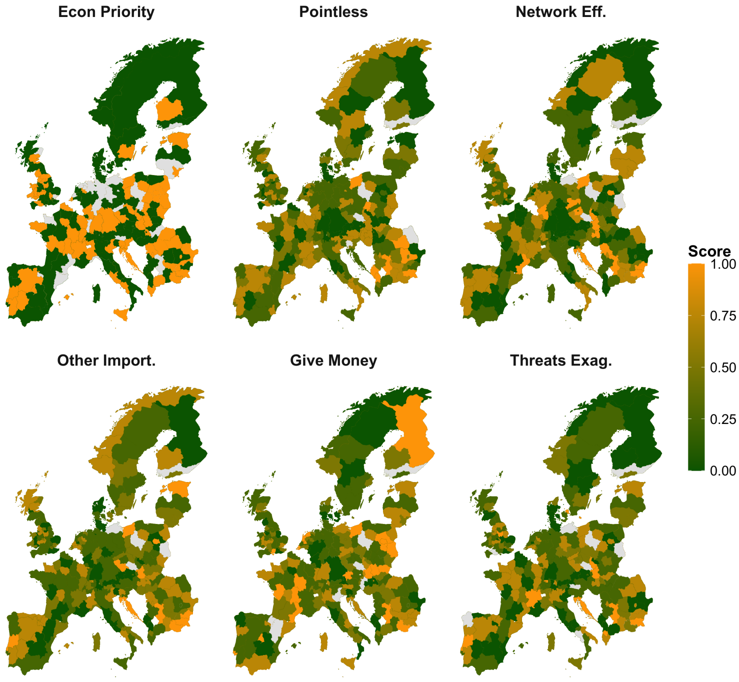


Figure 4: Distribution of the 6 environmental questions across Europe considering the average scores per region.

Since we have 6 different closely related questions for environmental attitudes, we construct a single Environmental Attitudes Index. To do so, we run a Principal Component Analysis (PCA) to combine all 6 variables. The first principal component explains 45% of the variance, and we take the first principal component as our Environmental Attitudes Index (see Table 3 in the Appendix for PC1 loadings). The index is continuous between 0 and 1, with 0 being more pro-environment, and 1 being less pro-environment. We find that the components of Western European countries behave more pro-environment (green), while generally Eastern European countries have less pro-environmental attitudes (yellow), as illustrated in Figure 5.

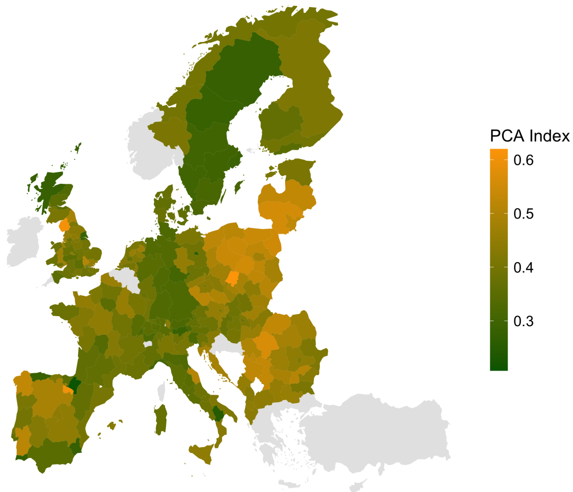


Figure 5: Environmental Index (PCA) across Europe considering the average scores per region.

4.3 Regression Model

Once we have all the variables together, we start our regression analysis. Our main specification is:

$$Y_{irc} = \alpha + \beta_1 Franc_r + \beta_2 Dom_r + X_i'\gamma + \delta_c + \varepsilon_{irc} \tag{2}$$

where Y_{irc} denotes the environmental attitude index for individual i in region r of country c (scaled 0-1), $Franc_r$ and Dom_r represent population-weighted exposure to Franciscan and Dominican Orders in region r measured in 1500, and X_i' is a vector of control variables, δ_c denotes country fixed effects, and ε_{irc} is the error term.

The control variables were taken from the EVS data, the same dataset as the environmental question variables. The vector of individual-level control variables, X_i', consists of: gender, age, education level, household income (PPP-adjusted), socio-economic status (ISEI), town size, political orientation (left–right scale), and religious denomination with dummy variables for Catholic and Protestant. We cluster standard errors at the NUTS level.

5 Results

5.1 Regressions and Robustness Checks

We started by running the first four regressions shown in Table 1 and found the following results. Model 1 is the simplest form of the regression, where we regressed the environmental index on the Franciscan exposure only. We see that the coefficient β_1 is negative and statistically significant. Because the environmental index ranges from 0 to 1, where lower values indicate greater pro-environmental behavior and higher values indicate less pro-environmental behavior, a negative coefficient indicates that the variable correlates with people being more pro-environmental. Thus, β_1 in Model 1 was in accordance with our initial hypothesis. In Model 2, adding the Dominican exposure decreases β_1. When we add country fixed effects in Model 3, β_1 changes its sign and loses its significance. In Model 4, when we add all the individual-level controls with the Dominican exposure and country fixed effects, β_1 becomes positive and statistically significant, which means that regions that had maximum Franciscan exposure had a 3.5 percentage points lower environmental index, opposite to our hypotesis. It is also woth noticing that many of the control variables are significant. For example, females are more pro-environmental. However, the coefficient of the Dominican exposure, β_2, is not statistically significant in any of the models we have analysed.

Table 1: Regression Results (SE clustered by region)

	Model 1 Franciscans	Model 2 Fran vs Dom	Model 3 Country FE	Model 4 Full + FE
Franciscan Exposure	-0.095^{***} (0.016)	-0.058^{*} (0.027)	0.004 (0.014)	0.035^{*} (0.015)
Dominican Exposure	–	−0.050 (0.027)	−0.013 (0.014)	−0.016 (0.014)
Female	–	–	–	-0.028^{***} (0.003)
Age	–	–	–	0.000^{**} (0.000)
Education	–	–	–	-0.000^{***} (0.000)
Income (PPP)	–	–	–	−0.002 (0.001)
Right-wing	–	–	–	0.007^{***} (0.001)
Town Size	–	–	–	-0.005^{**} (0.001)
Socio-Economic Index	–	–	–	-0.001^{***} (0.000)
Catholic	–	–	–	0.004 (0.004)
Protestant	–	–	–	−0.001 (0.005)
Observations	49,233	49,233	49,233	24,379
R^2	0.022	0.024	0.099	0.149
Adj. R^2	0.022	0.024	0.099	0.148

Notes: Standard errors in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

All the β_1 and β_2 coefficients from these regressions are plotted in Figure 6 (see Appendix). β_1 has a steady increase going from a negative to a positive sign as we move from Model 1 to Model 4, and finally becomes positive and statistically significant. β_2 is never significant.

Table 4 (see Appendix) reports the ìresults when we run regressions with each of the environmental questions separately, in order to assess whether there is one variable within the PCA index driving all of the results. We run the regression in Model 4 six times with one of the variables instead of the index. The values of β_1 and β_2 do not change drastically across these six regressions compared to Model 4 in Table 1. However, there aren't any significant results except for the coefficients of the variable indicating the collectivity effect, stating, "There is no point in doing what I can for the environment unless others do the same." The Franciscan and Dominican exposure coefficients produce significant results since they have contrasting theologies regarding this variable, where the Franciscan Order focused more on collectivity and the Dominican Order emphasized an individualistic view.

In Figure 7 (see Appendix), we run the main regression in Model 4 twenty-four times, each time excluding one country at a time to see whether there is one country driving the results. The plot on top shows the Dominican exposure coefficient, β_2, estimates across these regressions, and the bottom plot shows the Franciscan exposure coefficient, β_1, estimates. We see that there isn't much difference across countries. Only Switzerland and Italy have a slightly stronger effect on the results, but even excluding these countries, the results remain similar to those in Table 1. So the effects are homogeneous across Europe.

In Table 5 (see Appendix), we run the main regression again, this time excluding each control variable at a time for the sensitivity analysis. We see that the estimate of β_1 remains around 0.035 and is statistically significant across most of the regressions, except for one regression in which we exclude the town size control variable. This is an important result, as the Franciscans choose to locate themselves within urban areas to reach more people. Thus, not controlling for the town size would bias the coefficient of Franciscan exposure. This makes the town size a very important control, but apart from it, we could drop any other control and find very similar results to our full model.

In Figure 8 (see Appendix), we see the plotted regression coefficient estimates found in the sensitivity analysis in Table 5 by leaving out one control in each regression, the top plot showing estimates for β_2 and the bottom one showing estimates of β_1. We can clearly see that only the town size shifts the Franciscan exposure coefficient estimate. The Dominican exposure coefficient estimate also changes slightly when we remove income, political orientation, and town size, but none of the controls change the general direction of the results.

As additional robustness checks in Table 6 (see Appendix), we run subsample regressions with the full model, including all controls, but keeping only Catholic individuals in the first subsample. The Catholicism of individuals is determined from the same EVS data by asking the question "Which religious denomination do you belong to?" and taking into account only those who answered with the option "Roman Catholic". Considering only Catholics, the sample size drops significantly to 5899, and we don't have any significant results. In the second subsample, we exclude the observations from Germany to only keep NUTS-2 regions, to account for the lack of granularity in Germany. However, the results stay the same as in the main model.

Lastly, to account for possible selection bias into answering the survey questions about the control variables, We run the 4 models in the same, final àsample. Results stay similar (see figure 9 in the Appendix).

5.2 Possible Mechanism

Since we found contrasting results to what we were expecting to find, we find it important to propose a possible mechanism that might be driving these results, which could be used for further hypotheses and research. We have already ruled out all misleading mechanisms that could have come from the control variables, such as education, income, and political views, and from country fixed effects. For example, one could have claimed that Dominican exposure also meant a higher education level since they promoted teaching, which could have led to more pro-environmental attitudes. However, we do control for education level. Thus, the mechanism should be a deep, slow-moving, cultural or institutional effect.

We propose that the mechanism could lie in the opposite beliefs about taking collective action within the Franciscan versus Dominican theologies. The Franciscan theology prioritizes humility, acceptance of circumstances, poverty, and scarcity for the sake of imitating Christ, thus valuing moral purity over institutional action. The lack of desire for collective intervention might have shown itself through having less pro-environmental attitudes, which could also be observed in our Table 4 results (specifically beliefs about individual efforts being pointless unless others do the same, and environmental threats being exaggerated). We than theoreticize that Franciscans are not likely anti-nature, but rather anti-intervention. The Dominican theology, on the other hand, highlights persuasion and institutional coordination. Their willingness to take reasoned collective action and enforce norms could have translated into more modern pro-environmental attitudes. Their sense of obligation also possibly played a role in the context of the environment, and they might have had greater trust in institutions for collective solutions against environmental threats.

6 Conclusion

In this research, we analyzed whether the exposure to the two major Mendicant Orders of the Middle Ages, Franciscan and Dominican monasteries, has an effect on contemporary environmental attitudes. We constructed an exposure measure by combining the geodataset of monastery locations with historical population data, and we constructed an environmental attitudes index using the European Values Study survey data on individual environmental questions. We ran regressions controlling for various individual-level characteristics and country fixed effects. Through this, we have found that higher Franciscan exposure is associated with less pro-environmental attitudes. More specifically, in our main regression model, β_1 is 0.035 and statistically significant. Through various robustness checks, we have found that the results are quite robust and homogeneous across Europe and persist even when we exclude certain controls or countries.

We also proposed a possible mechanism that could be driving these results, which is the contrasting beliefs about collective action within the Franciscan and Dominican theologies. The Franciscan theology of humility and acceptance of circumstances might have led to less pro-environmental attitudes, while the Dominican theology of persuasion and institutional coordination might have led to more pro-environmental attitudes. The results might have been driven by underlying beliefs about whether environmental threats are exaggerated and whether individual efforts are pointless unless others do the same for the environment.

We can conclude our results by saying that long-run cultural and theological beliefs and monasteries can have a significant influence on contemporary attitudes toward the environment. The results of this research are quite interesting and surprising, but we cannot help but mention the limitations of our research and the need for further research.

7 Limitations and Further Research

One main limitation of our research is the Mapping Past Societies (MAPS) dataset, which contains foundation centuries only for Dominican monasteries, but not the specific dates, and it doesn't contain closing dates for any monasteries. Thus, we had to assume that all the monasteries were active throughout the period of 1200-1500, and so our exposure measure treats monastery presence as time-invariant.

The main extension that we propose is to use data on the foundation and closing dates of the monasteries to construct a more accurate exposure measure. We propose to use another existing dataset from Boranbay and Guerriero (2019) specifically for the Franciscan monasteries. They use a novel dataset which contains 2980 observations (while only 994 observations in MAPS) with specific foundation dates and monastery names for Franciscan monasteries. However, there are still no closing dates available in this dataset. Still, this would allow us to construct a more accurate exposure measure for the Franciscan monasteries.

Another limitation is that our specification is purely historical and correlational. We cannot fully rule out the deep unobserved regional culture, so we cannot make any causal claims without a proper experiment. The ideal experiment would be to randomly have a Franciscan monastery and then see what happens to these people who live there over time relative to everybody else. But for that we need the random assignment of monasteries. There is also a problem with the fact that there are no Franciscan monasteries in some NUTS regions in Northern Europe, and also no Dominican monasteries in some NUTS regions in Eastern Europe, so we cannot compare the effects of the two orders in those regions. A possible extentiuon is thus to exclude the regions that do not have either of the monasteries.

Another major issue is that we rely on data from 800-500 years ago and assume that the influence of the monasteries has persisted until today on the local populations, but don't know whether the people in the EVS dataset who live there now are the ones whose ancestors were effected or exposed to the monasteries in those regions during the Middle Ages. So as an extension, we would have to think about migration flows, to see how much of the population is actually exposed to the monastic influence.

Another possible consideration is whether all of the controls are indeed good controls. Some of the effect of the exposure might be biased because of the control variables, and so we would call these bad controls. This is probably not the case for the controls age and gender, but it might be the case for all the other controls, such as education, income, political orientation, and town size.A possible extention is thus to consider the controls more carefully, and run different combinations of Model 4.

As other possible extensions, we could check for the choice of monastery locations. It is possible (and expected) that monasteries were not randomly located across Europe. In this scenario, we could construct insturumental variables for the monastery locations, such as the distance to the nearest city or trade route, use additional controls, or explicitly model the endogenous placement to improve the causal identification and remove the endogeneity problem.

We could also construct a distance-weighted or decay-based exposure measure. Instead of assuming that everyone within a 25 km radius of the monastery is exposed to the theology of that monastery in the same way, we could assume that the influence decays with distance and people are less exposed the further they are.

We would also like to test the effects of monasteries on other outcomes, such as climate policy support, trust in institutions, and collective action. By using other datasets, we could extend our research to other countries outside of Europe, such as Latin America, where the influence of the Catholic Church is also quite strong. For example, Waldinger (2017) uses a Mexican dataset to show the long-term effects of missionary orders, such as the Franciscans, Dominicans, and the Jesuits, on outcomes such as education and Catholicism.

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9 Appendix

Environmental Attitude Survey Questions

Table 2 reports the exact wording and coding of the six environmental attitude questions from the European Values Study (EVS) used in the construction of the Environmental Attitudes Index. Five of the questions with possible answers agree strongly, agree, neither agree nor disagree, disagree, disagree strongly, are originally measured on a five-point Likert scale (1, 2, 3, 4, 5). The other one question with possible answers to choose between two statements "protecting the environment priority, even if slower economic growth and loss of jobs" vs. "economic growth and creating jobs priority, even if environment suffers" has two answers, scaled 1 or 5. In this case, choosing protecting the environment is coded as 1, and choosing economic growth is coded as 5.

We rescale all variables to a common interval between 0 and 1 and treat them as continuous in the analysis to make it easy to combine all questions into our PCA on a 0-1 scale. Due to the discrete nature of the original response categories, the rescaled variables only assume the values 0.0, 0.2, 0.4, 0.6, 0.8, and 1.0. Higher values indicate less pro-environmental attitudes, while lower values indicate more pro-environmental attitudes. The rescaling is implemented using a linear transformation of each variable to the unit interval.

Table 2: Environmental Attitude Survey Questions (European Values Study)

Survey Question (EVS)	**Scale**
Choosing between "protecting the environment priority, even if slower economic growth and loss of jobs" vs. "economic growth and creating jobs priority, even if environment suffers"	only 1 and 5 rescaled to [0,1]
"I would give part of my income if I were certain that the money would be used to prevent environmental pollution."	1, 2, 3, 4, 5 rescaled to [0,1]
"It is just too difficult for someone like me to do much about the environment."	1, 2, 3, 4, 5 rescaled to [0,1]
"There are more important things to do in life than protect the environment."	1, 2, 3, 4, 5 rescaled to [0,1]
"There is no point in doing what I can for the environment unless others do the same."	1, 2, 3, 4, 5 rescaled to [0,1]
"Many of the claims about environmental threats are exaggerated."	1, 2, 3, 4, 5 rescaled to [0,1]

Table 3: Environmental Index: PC1 Loadings

	PC1 Loading
envir_econ_priority	0.855
envir_efforts_pointless	-0.287
envir_other_importances	0.498
envir_network_effect	0.200
envir_threats_exaggerated	0.510
envir_protection_money	0.085

	Value
SS Loadings (PC1)	1.368
Proportion of Variance Explained	0.228

Regression Results and Robustness Checks

Figure 6: Impact of Monastic Orders (Clustered SEs)
95% CI clustered by region. Crossing 0 = Not Significant

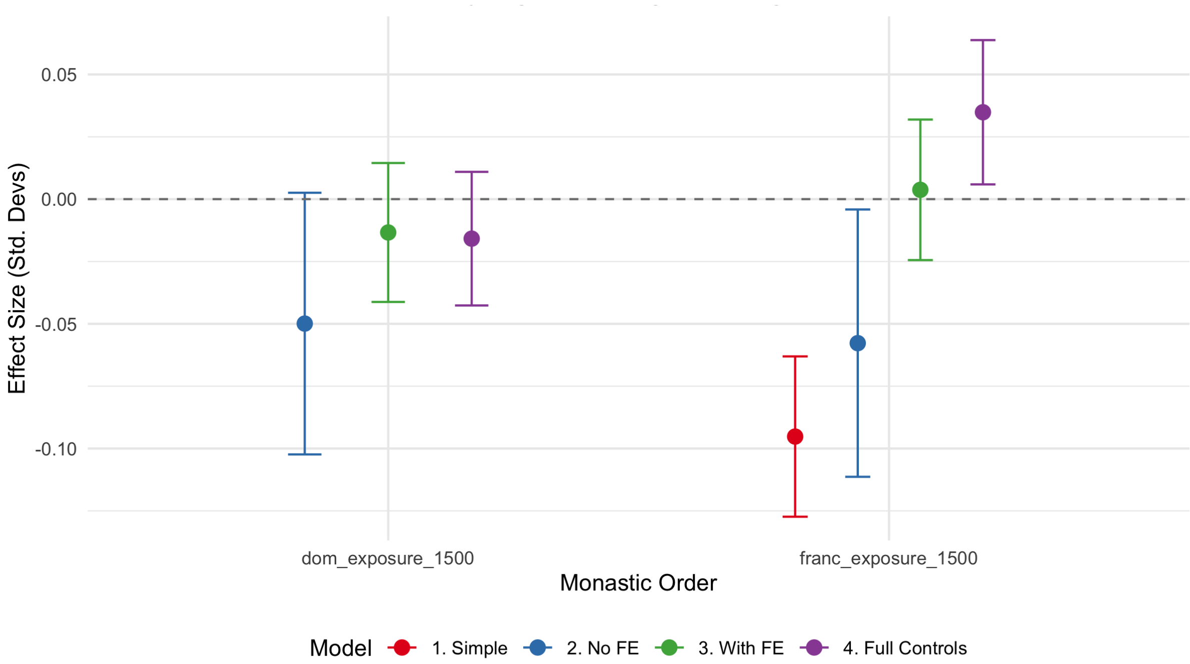

Table 4: Individual Environmental Questions (Clustered SEs)

	(1) Econ Priority	(2) Give Money	(3) Too difficult	(4) Other Import.	(5) Collective	(6) Threats Exag.
Franciscan Exp.	0.052	0.016	0.034	0.030	0.064**	0.037
	(0.038)	(0.021)	(0.021)	(0.019)	(0.021)	(0.021)
Dominican Exp.	-0.020	-0.026	-0.014	-0.018	-0.039*	-0.017
	(0.037)	(0.021)	(0.018)	(0.016)	(0.016)	(0.015)
Gender (Female)	-0.027***	-0.002	-0.023***	-0.040***	-0.028***	-0.047***
	(0.007)	(0.004)	(0.004)	(0.004)	(0.005)	(0.004)
Age	0.001**	0.001***	0.002***	0.000*	0.001***	0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Education	-0.000**	-0.000***	-0.000***	-0.000***	-0.000***	-0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Income (PPP)	-0.002	-0.004*	-0.011***	-0.004**	-0.010***	-0.004**
	(0.002)	(0.002)	(0.002)	(0.001)	(0.002)	(0.001)
Socio-Econ. Ind.	-0.002***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Town Size	-0.003	-0.006**	-0.001	-0.004*	-0.009***	-0.012***
	(0.004)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Polit. Orient. (R)	0.011***	0.006***	0.003**	0.004***	0.005***	0.010***
	(0.003)	(0.002)	(0.001)	(0.001)	(0.001)	(0.002)
Catholic	0.007	0.005	0.011*	0.010	0.017*	0.007
	(0.009)	(0.007)	(0.006)	(0.005)	(0.007)	(0.007)
Protestant	-0.003	0.003	-0.007	-0.017**	-0.004	0.008
	(0.013)	(0.012)	(0.006)	(0.005)	(0.006)	(0.008)
Num. Obs.	25,427	27,212	27,344	27,370	27,343	26,823
R^2	0.095	0.121	0.140	0.141	0.125	0.121
Adj. R^2	0.094	0.120	0.139	0.139	0.124	0.120

Notes: Standard errors in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 5: Sensitivity Analysis: Coefficients when Dropping Controls

	Full Model	Excl. Gender	Excl. Age	Excl. Education	Excl. Income	Excl. Politics	Excl. Town Size	Excl. Socio-Econ. Index	Excl. Religion
Franc. Exp.	0.035*	0.036*	0.035*	0.035*	0.032*	0.033*	0.012	0.039**	0.034*
Dom. Exp.	-0.016	-0.016	-0.015	-0.017	-0.008	-0.010	-0.010	-0.020	-0.016
Num. Obs.	24,379	24,379	24,434	24,438	26,795	33,560	28,220	27,286	24,379
R^2	0.149	0.144	0.149	0.147	0.143	0.135	0.141	0.148	0.149
Adj. R^2	0.148	0.143	0.147	0.145	0.142	0.134	0.139	0.147	0.148

Notes: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Figure 7: Does one country drive the results?
Coefficients when excluding each country one at a time (Clustered 95% CI)

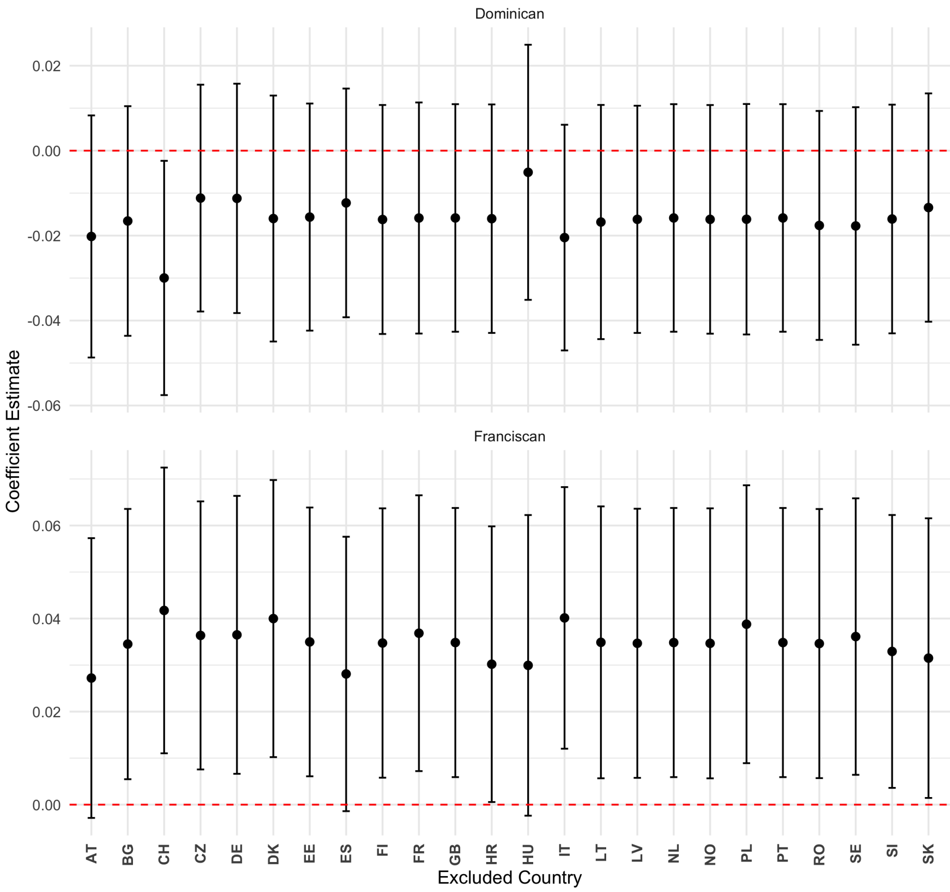

Table 6: Robustness Checks: Subsamples

	Main Model (All)	Catholics Only	Exclude Germany
Franciscan Exposure	0.035*	0.023	0.037*
	(0.015)	(0.021)	(0.015)
Dominican Exposure	-0.016	0.004	-0.011
	(0.014)	(0.018)	(0.014)
Num. Obs.	24,379	5,899	23,182
R^2	0.149	0.132	0.146
Adj. R^2	0.148	0.126	0.145

Notes: Standard errors in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Figure 8: Sensitivity Analysis: Leave-One-Control-Out
Does dropping a specific control variable change the results?

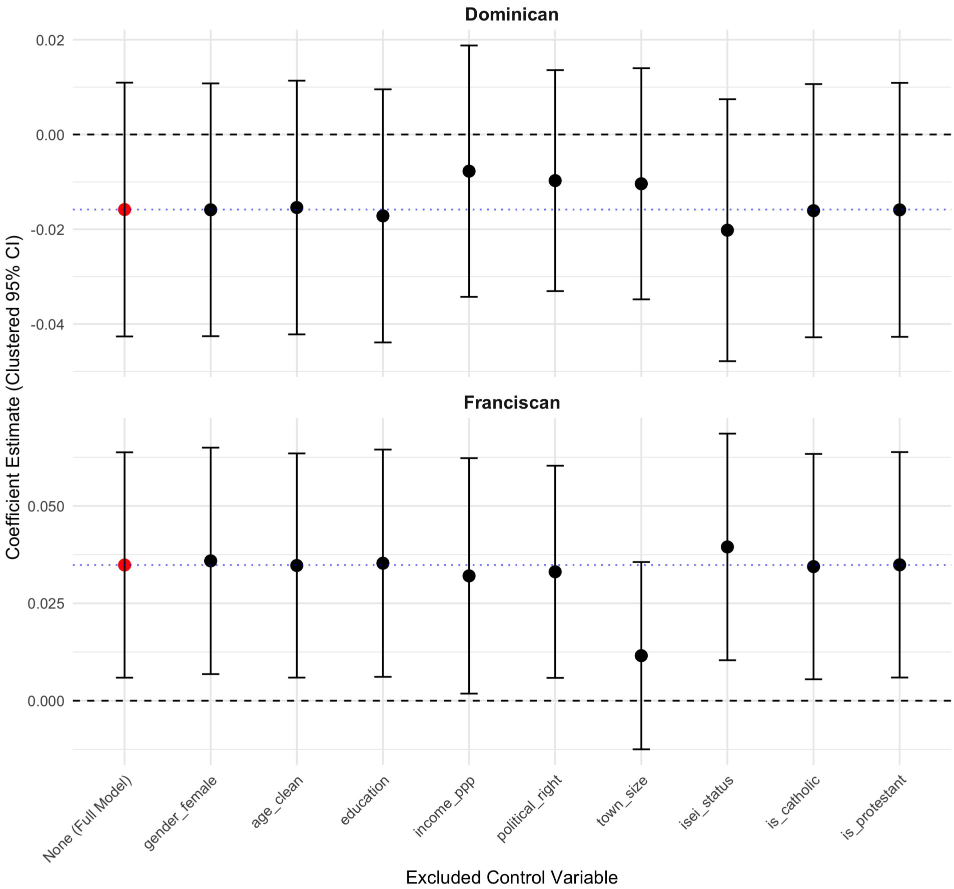

Figure 9: 4 Regression models in the same sample
Are results affected by self selection into answering the control variables questions?